

Design and experimental validation of a Marsupial Long-Endurance UAV-UGV system

Simón Martínez-Rozas¹ and David Alejo² and José Javier Carpio³ and Luis Merino³ and Fernando Caballero³

Abstract—Over the last decades, we have experienced an exponential growth in the use of Unmanned Aerial Vehicles (UAVs) for inspection operations. However, the reduced flight autonomy of electric powered UAVs can limit their applicability in long duration operations. In this paper, we propose a marsupial system composed of a UAV and a Unmanned Ground Vehicle (UGV). The system is made with off-the-shelf components that can greatly extend the flight duration of the UAV by powering it from the UGV using a tether, and reducing the mobility limitations by the coordinated motion of the system. We present the hardware design, making emphasis in the decisions made and presenting different alternatives depending on the application. Moreover, the software architecture for operating the device, based on the Robotic Operating System (ROS) is presented. Finally, the system is validated in experiments lasting more than one hour.

Keywords - Tethered robots, Long-endurance flight, Marsupial system

I. INTRODUCTION

In this paper, we focus on the use of a UAV for civil infrastructure inspection. In this area, UAVs have become a widespread and useful tool for applications as diverse as structure assessment on buildings and bridges [1], power line inspection, fire monitoring and extinguishing [2], [3] and even sewer inspection [4].

However, there is a strong limitation that may prevent an electric-powered UAV to perform an in-depth inspection of structures, which is its flight time. Indeed, most commercial platforms exhibit a flight duration well under half an hour, which is usually measured in ideal circumstances, without any additional payload (i.e. computing devices, sensors). Also, the altitude and air temperature and density can affect the flight duration of UAVs.

Some strategies can be employed to extend the flight duration such as manual battery swapping, using swarms of drones, and battery swapping stations. However, they came at a cost of reducing the automation of the system, increasing its complexity [5] or being still at an experimental level [6]. Another alternative is to use tethered UAVs fixed to a base station that supplies its power [7]. In fact, tethered

This work was supported by the grants INSERTION PID2021-127648OB-C31 and RATEC PDC2022-133643-C21, funded by MCIN/AEI/10.13039/501100011033 and “European Union NextGenerationEU/PRTR”.

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Fig. 1. Proposed marsupial system for long-endurance inspection missions.

UAVs have gathered the attention of the community [8] with applications as diverse as collaborative load transportation [9] and airborne wind energy [10]. However, the use of a tether fixed to a location may limit the UAV range to a great extent, preventing it to safely reach some areas of interest. Finally, Unmanned Ground Vehicles (UGVs), while capable of operating during longer periods, are not suited to this task, as they cannot reach inspection points located in high areas.

A marsupial multi-robot system [11] is a system in which one robot carries and can deploy one or several other robots. Some notable examples are the Oxpeter system [12], used for mining inspection. Also, the Team CSIRO Data61 [13] reached second place in the DARPA 2021 Subterranean Challenge [14] with a Marsupial System. Last but not least, the Mars 2020 rover and helicopter [15] which is being used for planetary exploration on Mars. In this system, the helicopter augments the capabilities of the rover, exploring large areas faster than the rover, providing reconnaissance on target locations and safe to traverse routes.

In this paper, we aim to devise, design and implement a marsupial robotic system (see Fig. 1) able to perform long-endurance inspection and surveillance tasks such as the assessment of the structural health of civil buildings. In our system, the UAV is linked to the UGV by means of a tether, which is used to provide the UAV with power to operate during long-endurance missions. Our design differs to some of the designs in the literature in that it uses off-the-shelf systems on the UAV, the UGV and the tether system. This feature makes it easy to be replicated and redesigned by the

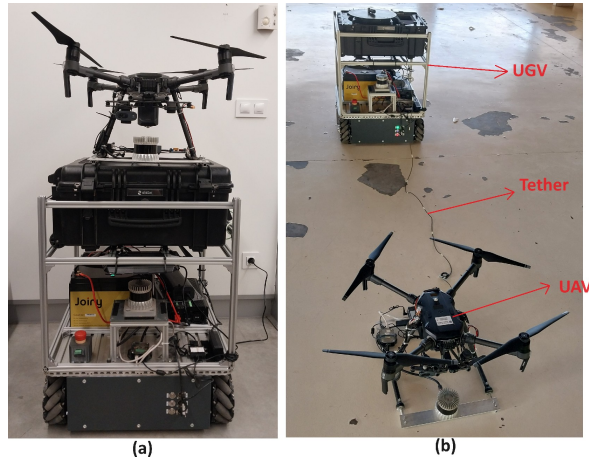


Fig. 2. Marsupial System, UAV tied to UGV. (a) Complete system previous a mission. (b) System in an initial position to begin the UAV takeoff.

community. Moreover, almost all of the software used in the system is available under free software licenses, running under Robotic Operating System (ROS). With this design, we have managed to perform autonomous long-endurance inspection missions with LiDAR based localization without the help of any external localization systems.

The rest of the paper is structured as follows: Section II describes the system hardware in relation to UGV, UAV and tether system. The design of the power system for the UAV is presented in Section III. The software architecture of the system is presented in Section IV. The experimental tests are discussed in Section V. Finally, the conclusions and future research directions are detailed in Section VI.

II. DESCRIPTION OF THE SYSTEM

We have designed a robotic system to carry out long-term inspection tasks. This system is composed of a UGV, a UAV, and a tether system (see Fig. 2). The main idea is to increase the flight time of the UAV by providing power from the UGV through the tether. Next, the main components of the platforms are described:

A. Unmanned ground vehicle - UGV

We use a ARCO ground platform from ID-Mind¹ [16], which is a 4-wheeled holonomic robot with an independent traction system (see Fig. 3a) that can work in holonomic or differential configurations. The robot is able to navigate at a maximum speed of 0.8 m/s. It carries two LiFePo 24V batteries (for motors and electronics), with an autonomy of 3 hours of operation. ARCO is a robot designed for delivery of heavy loads in factories, making us able to equip it with a specialized structure for the tether system directly on the platform, as shown in Fig. 3b. The platform has a maximum payload of over 100 kg, which is well above our requirements.

To carry out the task we have properly equipped ARCO with the necessary components to reach the marsupial configuration (see Fig. 4). We list in detail the components below:

¹<https://www.idmind.pt/arco>

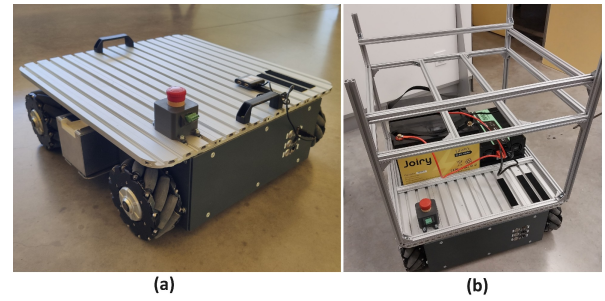


Fig. 3. ARCO, a Unmanned Ground Robot able to work in holonomic or differential mode. (a) Basic ARCO configuration. (b) ARCO configuration with auxiliary structure for tether system.

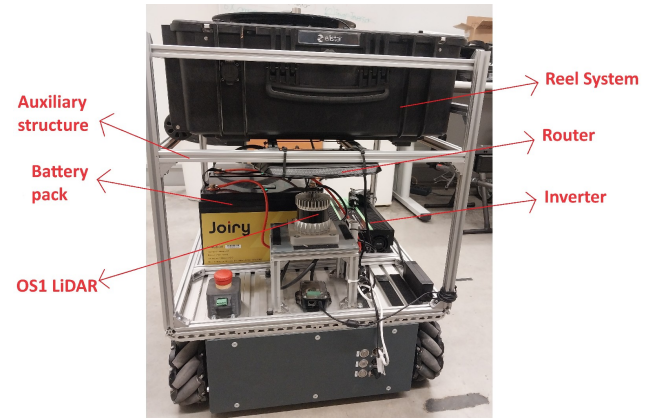


Fig. 4. ARCO robot integrates with sensor, power system, and auxiliary structure to support tether system and place devices

- Onboard Computer: MSI CubiN 8GL-001BEU with an Intel i5 processor placed inside of the platform.
- Inertial Measurement Unit (IMU): Sparkfun Razor 9 DoF IMU placed inside of the platform. It is used for mapping and localization modules.
- LiDAR: A 3D OS1-16 Ouster LiDAR, located on top of the robot. The LiDAR is able to provide 320k 3D points per second, ranging from 0.2 to 120 meters. The information gathered by this sensor is used for localization, mapping and obstacle avoidance.
- Auxiliary structure: Designed primarily to position the tether system at the top of the UGV. It is built from Boch RexRoth aluminum profiles.
- Battery pack: this is part of the tethered power system. Two LiFePO4 batteries store the energy that is provided to the UAV to increase its flight time.
- Inverter: this is part of the tethered power system. It is responsible for converting the DC energy from the batteries into AC energy for the tether system.
- Reel System: It is placed on top of the auxiliary structure. It is responsible for connecting the Power System with the UAV.
- Router: The marsupial system is equipped with a NIGHTHAWK NetGear AX5400 Router. We configured the local network, so that UGV is connected via Ethernet and the UAV via 5G WiFi.

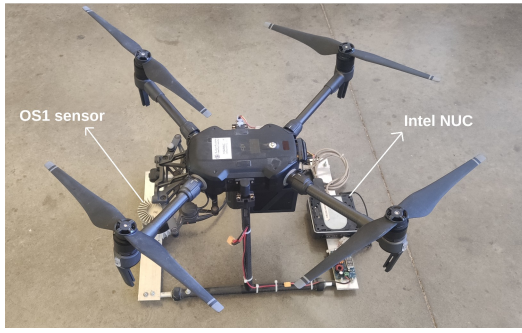


Fig. 5. DJI Matrice 210 V2 UAV.

B. Unmanned aerial vehicle

Regarding the aerial platform, the model used for the experiments is the DJI Matrice 210 V2 (M210) (see Fig. 5). We selected this platform because of the robustness of the hardware and software, the support that the provider offers, and its ROS support. The drone's structure was adapted in order to carry an i7 NUC computer and a LIDAR, as shown Fig. 5. The main sensors and systems onboard the UAV are:

- DJI Matrice onboard sensors: This includes a single GPS, an Inertial Measurement Unit (IMU) and a barometric altimeter.
- 3D LIDAR sensor: An OS1-16 Ouster LIDAR is used for the purpose of environment mapping, localization, and odometry.
- WiFi module: It is used to increase the robustness, range and performance of the WiFi connection at the UAV.

C. Tether System

We use the system LIGH-T V4 Tethered Station (see Fig. 6), provided by the company Elistair². We selected this system due to the support that provides the company and its compatibility with the DJI M210 V2. The cable of the system ends in an adapter, which is located in the drone battery compartment to provide energy to the drone. The output of this battery is 6S or 12S (battery in series) related to LiPo 3.7 V. In our case, we must provide to the drone with 22.8 V, so, we must power the drone in a 6S configuration, which means $6 \times 3.7 \text{ V} = 22.2 \text{ V}$. Finally, even though the UAV draws its power from the Elistair system, it is also equipped with a backup battery for the sake of safety.

The tether length is around 70 m. Besides, the system works with 110 – 220 VAC (50 – 60 Hz) voltage input, having a power consumption of 1200 W. This is the reason why we use an inverter, to convert from VDC from the battery pack to VAC needed for the system. The reel is connected to a DC motor, which can be configured to perform a constant tension to the cable ranging from 1N to 3N.

III. POWER SYSTEM REQUIREMENTS

In order to increase the flight time of the UAV, we must include in the marsupial robot a power system able to provide

²<https://elistair.com/>

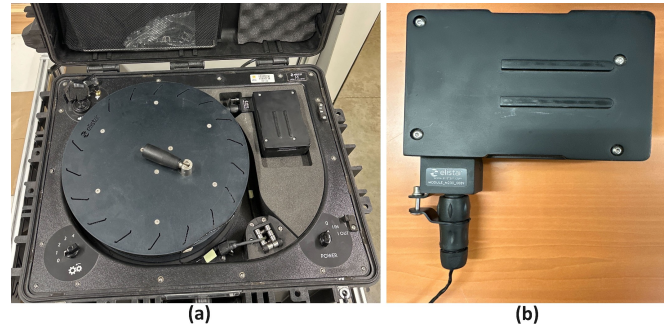


Fig. 6. (a) LIGH-T V4 Tethered Station, including switches, a reel for the cable, and the battery connector. (b) Detail of the battery connector.

energy for a long time. In this section, we detail how these components were designed and sized.

The requirement defined for a long-duration flight in a marsupial configuration is that the UAV must be capable of carrying out a one hour inspection task with 60% of remaining battery, thus exceeding two hours of flight duration.

Figure 7 represents the interconnection scheme of the power system, which consists of the following components:

- First, it is required that the inverter performs 220V/230V AC output to power the tether system. Besides, because of the limitation of the space to install the inverter on the ARCO, we must restrict the size. Under these considerations we selected the Green Cell® 2000W/4000W inverter from 12V DC to 220V/230V Pure Sine Wave., with dimensions: 370 x 180 x 80 mm.
- Second, taking into account that the inverter inputs are 12 V, batteries were considered in this operating range. We use LiFePo4 batteries given their safety in terms of thermal and chemical stability, which significantly reduces the risk of fire or explosion due to overload, short circuits, or damage. In addition, they have a long useful life, supporting a greater number of charge and discharge cycles (typically more than 2000 cycles) while maintaining a high capacity, which translates into a longer useful life compared to other battery technologies. Finally, they have a high energy density, sufficient for most high-power and long-life applications [17]. For these reasons, we selected the battery Joiry, 12.8 V, 150 Ah thus 1920 Wh), 328 x 172 x 214 mm. Considering the power consumption of the tether system (1200 W) we obtain a flight duration of 1 h and 36 min.

Given the demand of the tether system, in 1 hour 62.5% of the battery will be consumed ($1200/1920 \times 100$). To avoid this, and reach the 60% charge of the batteries defined as a requirement, we connect two batteries in parallel, for a total of 300 Ah. In this way, we can extend the theoretical flight duration to 3h and 12 min.

IV. SOFTWARE ARCHITECTURE

The marsupial software architecture is a set of modules that make use of the robot platforms, sensors, and communications device, described in the previous sections, to develop

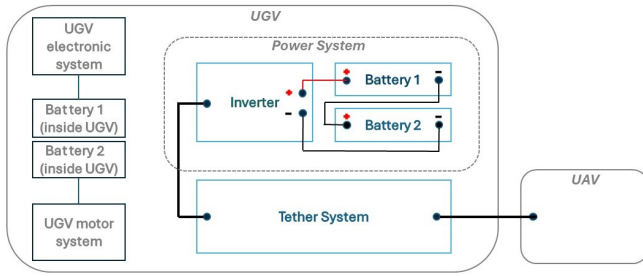


Fig. 7. Connection diagram of the power system onboard the UGV.

the required inspection task autonomously. The software used in the system is available under free licenses and operates under ROS noetic³ on Ubuntu 20.04. Figure 8 shows the main ROS nodes found in our system and the information flow between them. They are listed as follows:

A. Base station Software

- **Trajectory Planner:** We use a two-step planner in which a initial solution computed with a RRT* algorithm is then optimized with a non-linear optimizer that takes into account the kinematic and dynamic constraints and enhances the safety of the routes. We use the approach presented in [18] but with a taut tether.
- **Trajectory Coordination:** It coordinates the trajectory tracking on the UAV and UGV. We use a loosely-coupled solution in which new waypoints cannot be commanded until both systems reach the current one.

B. UGV Software

- **ROS Master:** A ROS Master Core is executed in the UGV computer. It is necessary to establish the communications between the ROS nodes of each computer.
- **ARCO driver:** This software interfaces with the hardware board of the robot, and offers standard ROS commands and services for easy integration of the robot with the rest of the system.
- **OS1 LiDAR driver:** It communicates the data generated by the sensor with ROS, so that the data is used by other modules, such as localization.
- **UGV localization:** This module uses a prior 3D map of the environment and the LiDAR data to localize the platform using Direct LiDAR Localization (DLL [19]).
- **Trajectory Tracker:** This module receives the trajectory computed by the base station and sends commands to ARCO driver module. We use a proportional control with saturation for maintaining the heading reference, then the speed tracking follows a trapezoidal profile, similar to the one presented in [3]. Please note that as the UGV platform is holonomic, we can send velocity commands in both the x and y axes of the UGV.

C. UAV Software

- **DJI driver:** Software that interfaces with the hardware board of the UAV, and offers standard ROS topics and

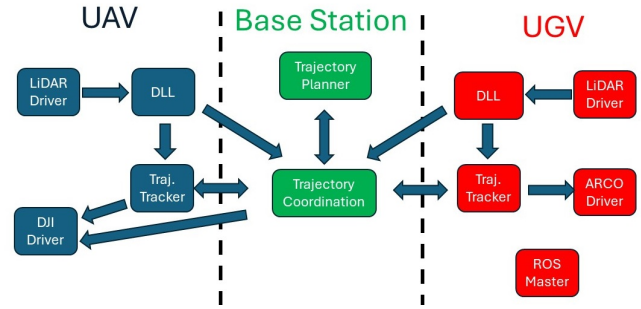


Fig. 8. Main software components of the Marsupial System to carry out autonomous missions.

services to access the internal DJI devices such as motor and sensors.

- **OS1 LiDAR driver:** Same as the UGV.
- **UAV Localization:** Same as the UGV.
- **Trajectory Tracker:** Similar to the one at the UGV, but adding a proportional height controller.

V. EXPERIMENTAL RESULTS

To test the capability of our system to extend the flight duration of the UAV by powering it from the UGV using a tether and its use in inspection tasks, we have executed field experiments which are presented in this section. Two different experiments were carried out at the Universidad Pablo de Olavide (see Fig. 9). First, we performed a flight endurance test at Building 45. Then, we performed an inspection simulation at the Old Theater, in which the inspection mission was executed in autonomous mode.

A. Endurance tests

We carried out a flight endurance experiment at the building 45 of the Pablo de Olavide University. The test was performed in manual mode and the platforms were controlled to get a total displacement of over one kilometer, mimicking the stop and go behavior in a inspection mission.

The experiment exceeded one hour duration, which allowed us to get information about the discharging rate of both the onboard UAV and the batteries on the UGV used for powering the UAV via the Elistair tether system. Figure 10 represents the battery percentage of the backup battery onboard the UAV. Note that the battery slowly discharges until stabilizing at 86%. This discharge is normal, as the battery discharges so that its voltage level matches the one provided by the Elistair system remaining stable afterwards.

Table I represents the level of the batteries powering the Elistair system and the onboard UAV battery used as a backup, for ease of comparison, during the experiment. As expected, the discharge rate of the batteries remains constant during the experiment, and the battery levels ended above 50%. Therefore, the battery draining of the system is slightly higher than expected (63%). This can be explained because some components also use these batteries, such as the LiDAR and the router. In spite of this, our platform can exceed two hours of flight duration.

³<https://www.ros.org/>



Fig. 9. Environments where the experiments were carried out. (a) Endurance tests at building 45. (b) Inspection mission at the Old Theater.

TABLE I
CHARGE OF THE BATTERIES, IN [%] OF THE DIFFERENT SYSTEMS AS
THE MISSION TIME PROGRESSES.

Battery Percentage					
Time [min]	0	15	30	45	60
Battery 1 [%]	100	88	76	64	52
Battery 2 [%]	100	89	77	66	54
Onboard Battery [%]	100	92	88	86	86

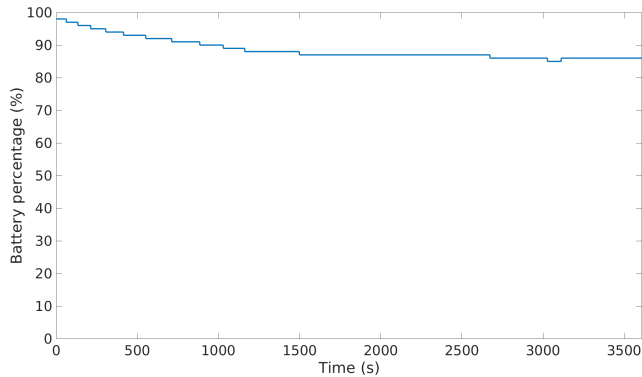


Fig. 10. Battery percentage on the backup battery onboard the UAV during the endurance tests.

B. Emulated Inspection Tests

Finally, to test the whole system in autonomous mode, we emulated a structural assessment inspection. Our marsupial system had to look for emulated defects in the Old Theatre building of the Universidad Pablo de Olavide. To emulate the defects, we disposed an array of twelve AR markers with a side of fifteen centimeters on the walls of the building (see Fig. 9b). We equipped the aerial platform with a high resolution camera to automatically detect the markers by using the ArUco library [20].

Figure 11 represents an obtained 3D map of the environment, which was used for localization and path planning purposes on both platforms. The goal of the experiment is to make an in-depth inspection of one of the walls of the building. To this end, a zig-zag exploration trajectory was

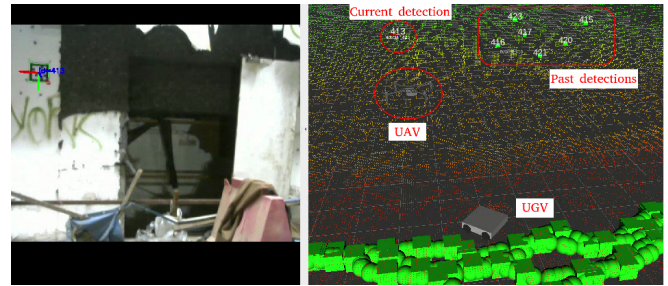


Fig. 11. (Left). Detection of a simulated defect and its related axes. (Right). At the same instant, the state of the UAV and UGV represented in the 3D map of the environment, past detections and trajectory performed by the UGV (green).

computed using our trajectory planner module (see Section IV). The great endurance of the platform allowed us to fly the trajectory three times without changing any batteries.

The trajectories flown in one of the experiments and the position of the UAV and UGV are also represented in Fig. 11. With the designed trajectories, the platforms were able to detect a all the simulated defects in the three experiments, which indicates the repeatability of the trajectories. Please refer to the video⁴ for more details of the experiments.

C. Lessons Learned

There are multitude of things necessary for a multi-robot experiment to be successful. In this section, we point out the some critical things we have learned from experimentation.

- **Be careful with the ordering when powering up the systems.** The main drawback of using ROS for multi-robot experimentation is the use of a centralized system for communication purposes, which is the ROS Master process. One solution could be to use a multi-master approach [21]. However, for the sake of simplicity, we use a ROS master process at the UGV computer which is powered on during the experimental session. Then, both the laptop and onboard UAV computer are

⁴<https://youtu.be/BLCVO0qSJZo>

configured to use this ROS master. Therefore, we always power up the UGV in first instance and its onboard router. Then, we use the laptop of the base station to check all the systems. Finally, the UAV computer is only powered up just before the beginning of a experiment.

- **Prepare and label several UAV batteries before the experiment.** Even though the energy consumption on the onboard UAV battery is low during the experiment (see Fig. 10). The DJI firmware checks the onboard batteries to be fully charged as a condition for powering up the system. Therefore, it is a good idea to have several spare UAV batteries for a long run experimental session.
- **Take care of the relative position between the UAV and UGV.** The LIGH-T V4 station has a mechanism that does a constant tension to the tether, keeping it taut. Experimentally, we found out that a tension of $1N$ is enough to keep it straight in most situations. The problem is that this tension should be handled by the control system of the UAV. When the tension is performed downwards it has a similar effect as increasing of the payload of the system. However, when the tension is performed laterally, the control system has to change the attitude of the UAV to compensate the disturbances. Therefore, it is better for the UAV to be as close as possible to the UGV to avoid lateral disturbances. Finally, we fix the tether as close to the center of gravity of the UAV as possible, preventing the appearance of torques.

VI. CONCLUSIONS

In this paper we have designed a marsupial UGV-UAV system for extended UAV autonomy. We show experimentally that, even though consuming more than expected, the platform can perform more than two hours long inspection missions, greatly extending original the flight time of the platform, which was of twenty-five minutes. However, the flight endurance of the platform can be further extended by adding additional batteries to the UGV, as we are far from its payload limits. Moreover, we described the software architecture of the platform that has enabled us to perform an emulated civil structure assessment autonomous mission. Future efforts will be devoted to performing long-range structure assessment tests with the system and to adapting the software of the system to ROS2, integrating it into the Aerostack 2 framework [22].

ACKNOWLEDGMENT

The authors would like to thank Sara Victoria Hernández Díaz for her management and invaluable help to find the experimental scenario, and for her infinite patience. Also, we would like to thank all the Service Robotics Laboratory staff for their unselfish help during the experiments.

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